Dr. Dilip Mahalanabis showed the efficacy of ORS in cholera cases among Bangladeshi refugees (1971-72). Further field trial during the cholera epidemic in Manipur attested to its efficacy, ORS has since saved the lives of millions of children around the world.

Look at the above information. What do you infer from this? Now you get the curiosity to know about ORS and its function. Don't you? In addition to this, let us know about some of the common medicines and how do they work.

In the normal healthy intestine, there is a continuous exchange of water through the intestinal wall. Up to 20 liters of water is secreted and very nearly as much is reabsorbed every 24 hours. This mechanism allows the absorption of soluble metabolites into the bloodstream from digested food. However when a person becomes sick, due to diarrhea, water is expelled and the body is not able to retain the liquid balance. This is called as 'dehydration'. It is not the diarrhea that kills, but the dehydration' resulting from the infection that kills. If more than 10% of the body's fluid is lost death occurs.

DO YOU KNOW?

UNICEF/WHO norms the O.R.S should be prepared as follows

S.No.	New ORS	gram/Litre	%	New ORS	mmol/litre
	Sodium chloride	2.6	12.683	Sodium	75
	Glucose, (anhydrous)	13.5	65.854	Chloride	65
	Potassium chloride	1.5	7.317	Glucose, anhydrous	75
	Trisodium citrate, (dehydrate)	2.9	14.146	Potassium	20
				Citrate	10
	Total	20.5	100.00	Total Osmolarity	245

4. 1 Oral Rehydration Solution (ORS)

ORS (Oral Rehydration Solution) is a special combination of dry salt and sugar that is mixed with safe water. It can help to replace the fluids lost due to diarrhea, vomiting and sweat. In a state of diarrheal disease there is imbalance and much more water is secreted than reabsorbed causing a net loss to the body which can be as high as several liters a day. In addition to water loss, sodium and potassium are also lost.

Certain concentration of sodium (Na) is needed for proper functioning of the body. For example, only with adequate sodium concentration in the intestinal wall, water can be absorbed by it through a process known as osmosis. If there is inadequate salt in the intestinal wall the body will not be able to absorb water.

The saline bottle directly transfers water and sodium into the blood stream. However, for the saline water is administered through mouth, intestinal wall, is a not able to absorb neither water nor sodium. Dr. Dilip Mahalanabis found that if glucose (sugar) is added to the salt solution, then all the three- water, sodium and glucose are absorbed by the body.

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